

AGROPILOT

AI-Guided Field Force Intelligence

Complete Product Feature Document

Syngenta IITM Hackathon 2026 · Prototype Phase · May 2026

Platform	Mobile App (Field Reps) + Web Console (Managers)
Total Pages	17 screens across two user roles
Core Technology	GraphRAG · LangGraph · Claude AI · Neo4j · FastAPI
Season	Rabi 2025-26 (October 2025 – April 2026)
Target Users	Field Sales Representatives + Territory Managers

What is AgroPilot?

AgroPilot is an AI-powered field operations co-pilot built for Syngenta's field sales force in India. It replaces gut-feel visit planning and routine rotations with data-driven, real-time recommendations — telling each field representative exactly who to visit today, what to say when they arrive, and flagging market signals that demand immediate action.

This document describes every screen and every feature in the application. It is written for team members, evaluators, and anyone who needs to understand what AgroPilot does and why each feature exists.

The Problem Being Solved

Syngenta's field teams visit retailers and farmers daily, but their planning is largely based on habit and experience — visit the same retailers in the same order, week after week. This causes four specific problems that AgroPilot directly solves:

Missed timing windows

Crops have narrow biological windows — for example, a wheat fungicide must be applied during the heading stage or it is ineffective. Fixed visit rotations often miss these windows.

Ignored inventory signals

A retailer running out of stock is a crisis that needs a visit today, not in two weeks when the rep's rotation brings them back.

Slow response to market signals

A competitor promotional push, a pest outbreak, or a demand surge requires a fast field response — not a response based on last month's plan.

No learning from outcomes

When a rep makes a sale or gets rejected, that information should improve future recommendations. Today it is lost in notebooks or spreadsheets.

Who Uses This App

User	Device	What They Do	What They See
Field Rep	Android smartphone (offline capable)	Visits retailers and farmers daily across their territory	Full mobile app: all 11 rep screens
Territory Manager	Desktop / laptop browser	Oversees multiple reps, tracks KPIs and alerts	Web console: 5 manager screens

Key Design Principles

Explainability first

Every AI recommendation shows the reason behind it in plain language. Reps are not expected to blindly follow suggestions — they need to understand and communicate the logic.

Offline by default

Rural India has unreliable connectivity. Every critical feature works without an internet connection. Data syncs automatically when connectivity returns.

Mobile-first

The rep app is designed for one-handed use on a mid-range Android phone, in sunlight, between visits. Large tap targets, high contrast, minimal typing.

Rep-managed profiles

There is no separate login for farmers or retailers. The field rep creates and manages all profiles. This reflects the reality of rural India where the rep is the primary relationship owner.

What Is in This Document

PART A — Field Representative App (12 screens)

Page 1 — Dashboard / Morning Briefing

Daily command center — priority visit list, AI field score, crop windows

Page 2 — Route Planning

Smart map — AI-optimized visit sequence with manual override

Page 3 — Visit Copilot

Point-of-visit screen — AI pitch, stock status, talking points

Page 4 — Log Visit Outcome

Offline-capable outcome logging — closes the AI feedback loop

Page 5 — Alerts Feed

Proactive anomaly and opportunity detection across the territory

Page 6 — Crop Scanner + Disease Diagnosis

Camera-based crop diagnosis with Syngenta product recommendation

Page 7 — AI Chatbot Assistant

Voice and text co-pilot — always accessible from any screen

Page 7b — Reasoning Graph

Explainability screen — interactive evidence graph behind every AI recommendation

Page 8 — Yield Loss / ROI Calculator

Economic argument tool — shows farmer the cost of not acting

Page 9 — Farmer Profile

Rep-managed farmer records — crops, history, engagement

Page 10 — Retailer Profile

Rep-managed retailer records — stock, visits, competitor notes

Page 11 — Offline Mode and Sync

System-wide offline layer — pre-caching, queuing, auto-sync

PART B — Manager Console (5 screens)

Page 12 — Territory Heatmap

Visual coverage map — gaps, demand signals, competitor activity

Page 13 — Revenue and KPI Dashboard

The three success metrics from the problem statement

Page 14 — Rep Performance Tracker

Individual scorecards, leaderboard, AI usage rate

Page 15 — Alert Management

Cross-territory alert feed with assignment and resolution tracking

Page 16 — Campaign Performance

Digital funnel + field conversion linked together

Field Representative Mobile App

The rep app runs on Android smartphones and is designed for daily field use — often outdoors, sometimes without internet. All 12 screens are described below with their complete feature sets.

■ Page
1

Dashboard / Morning Briefing

FIELD REP ·
MOBILE

The Dashboard is the first screen a rep sees when they open the app each morning. It is designed to answer one question in under 30 seconds: what should I do today and why? Everything is pre-computed overnight by the AI so the rep can act immediately.

This screen covers both retailers AND farmers in the priority visit list. The highest-value action on any given day might be visiting a large farmer directly, not just a retailer — the AI treats both equally.

AI Field Score and Daily Summary

AI Field Score (0–100)

A single number summarising today's urgency across the entire territory. It is computed from five signals: inventory gaps at retailers, days since each retailer or farmer was last visited, proximity to critical crop growth stages, active digital campaign alignment, and any open anomaly alerts. A score above 80 means the rep has urgent actions today. Below 40 means it is a lighter day.

Daily summary strip

Sits below the score — shows at a glance: number of urgent visits, number of stockout alerts, and whether any crop biological window is closing this week.

Priority Visit List

AI-ranked visit cards

Each card in the list represents one retailer or farmer to visit today. They are ranked by a priority score that weighs inventory risk, crop stage urgency, visit gap, digital campaign warmth, and weather signals. The rep sees the ranked list and can act on it immediately — no manual planning required.

Plain-language AI reason

Every card shows exactly why this visit is recommended today, in one sentence the rep can understand and repeat to the retailer or farmer. For example: 'Topik 15 WP is out of stock here, and wheat heading stage opens in three days in this tehsil.' This satisfies the explainability requirement from the problem statement.

Urgency badge

Each card is tagged High, Medium, or Low. High means visit today. Medium means visit this week. Low means routine — can be deprioritised if time runs short.

Visit gap indicator

Shows how many days since the last visit to this retailer or farmer. This is shown as a coverage gap signal — not as a streak or reward — because the goal is ensuring no retailer or farmer is neglected, not encouraging repeated visits to the same place.

Distance from current location

Distance in kilometres shown on each card so the rep can make practical decisions about sequencing. The AI balances priority with geography — it will not send a rep across the district for a low-urgency visit when a high-urgency one is nearby.

Crop Window Countdown Strip

Biological stage indicators per crop

A horizontal strip showing all crops present in the territory — Wheat, Mustard, Chickpea, Potato — with their current growth stage and a countdown in days to the next critical window. For example: 'Wheat — Heading stage — 3 days to fungicide window closing.' This is agronomically critical context that drives the priority scoring.

Weather Strip

Tehsil-level weather summary

Shows today's temperature, conditions, and rain risk for the rep's territory. Rain is directly relevant — fungicide application is ineffective before rain, so the rep needs to know if planned recommendations should be brought forward or postponed.

Quick Access

Crop Scanner shortcut

Opens the Crop Scanner directly from the dashboard without navigating through menus. Useful when a rep hears about a disease problem on the way to a visit.

AI Chatbot bubble

A floating button gives access to the AI assistant from any screen, including the dashboard. The rep can ask 'What should I pitch at my first stop today?' before leaving the vehicle.

Offline indicator

A small dot in the top bar — green when connected, amber when offline. When offline, the dashboard loads from the morning cache. All data is pre-downloaded at app open.



The Route Planning screen helps the rep plan the most efficient visit sequence for the day. The AI does not just find the shortest route — it finds the route that balances geographic efficiency with visit priority so the most urgent stops are not sacrificed for convenience.

Territory map with priority pins

Shows all retailers and farmers in the rep's territory as map pins. Red pins are urgent visits, amber are moderate, green are routine. Each pin is numbered in the AI's recommended visit order so the rep sees immediately which stop is first, second, and so on.

AI route optimisation

The AI computes the most efficient sequence considering both the priority score of each stop and the travel distance between them. It will not route a rep past a high-urgency stockout retailer on the way to a low-urgency routine call.

Manual reorder with trade-off explanation

The rep can drag and reorder stops manually. When they do, the app shows what they are giving up — for example, 'By moving this stop to fourth, the out-of-stock retailer in Baramati will not be visited until the afternoon when the campaign window may have passed.'

Filter toggle

The rep can filter the map to show only retailers, only farmers, or both. Useful when planning a farmer-only day or a retailer stockcheck run.

Route summary

A panel at the bottom of the screen shows total kilometres, estimated travel time, number of stops, and how many of those stops are high priority.

GPS arrival detection

When the rep's GPS shows they have arrived at a stop, the Visit Copilot for that specific retailer or farmer opens automatically. No manual navigation needed.

Emergency re-route

If a new urgent alert fires during the day — for example, a sudden stockout cluster is detected — one tap re-optimises the remaining route around the new emergency stop.

Offline map cache

Territory map tiles are downloaded every morning when the app opens with connectivity. The map works fully offline during the day.

The Visit Copilot is the most important screen in the application. It opens when the rep arrives at a retailer or farmer and gives them everything they need to have a productive, informed conversation — without having to remember anything from memory.

The AI recommendation on this screen is always shown with its reasoning. The rep must understand why a product is being suggested — not just what to pitch. This directly addresses the explainability requirement in the problem statement.

Retailer or farmer header

Name, location, tehsil, owner or contact name, and best visit time based on the rep's historical notes. Shown at the top so the rep can confirm they are at the right place.

Previous visit summary

Shows the date of the last visit, what product was discussed, and what outcome was logged. This prevents the rep from repeating the same pitch visit after visit, which is one of the core problems the product is solving.

AI recommendation block

Tells the rep exactly what product to lead with and why, in plain language. The reason is grounded in the current stock level at this retailer, the growth stages of nearby farmers' crops, any relevant digital campaign activity this week, and the weather forecast for the tehsil. All reasoning is visible, not hidden.

Live stock status

Shows current inventory for key SKUs at this retailer — Topik 15 WP, Score 250 EC, Actara 25 WG, Kavach 75 WP — with Out of Stock shown in red, low stock in amber, and healthy levels in green. Exact unit counts are shown.

Nearby farmer list

Lists farmers in this tehsil with their crop type and current growth stage. This tells the rep which farmers the retailer is currently serving and why certain products are time-sensitive right now.

Agronomic talking points

Three to four bullet points the rep can use in conversation with the retailer or farmer, grounded in what is happening in the crop calendar this week. These are practical, specific, and easy to remember.

Competitor products section

The rep can log which competitor SKUs are visible at this retailer. This information feeds the territory-level competitor intelligence layer visible to managers.

Log Visit Outcome button

A prominent button at the bottom of the screen takes the rep to the outcome logging flow. The Visit Copilot and the outcome log are designed as a pair — one opens the other.

Every visit logged becomes training data for the AI. Over time, the system learns which types of recommendations led to sales, which did not, and why — and adjusts future recommendations accordingly. This is the outcome learning loop described in the problem statement.

Outcome selection

The rep selects one of four outcomes: Sale Made, Order Placed, No Purchase, or Follow-up Required. These are shown as large tap targets — easy to use with one thumb while standing at a counter.

Products discussed

Multi-select which Syngenta products were discussed during the visit. This is separate from the outcome — a rep might discuss three products and only get an order for one.

Quantity input

If an order was placed, the rep enters the quantity per SKU. This feeds into the inventory and demand signal system.

Competitor activity toggle

A quick toggle: did the rep observe any competitor products or promotional activity? If yes, a field opens to log which product and brand. This builds the competitor intelligence layer over time.

Follow-up scheduling

If the outcome was Follow-up Required, the rep picks a date. On that date, the relevant retailer or farmer automatically appears in the morning briefing's priority list.

Voice-to-text notes

A free-text notes field with voice dictation. The rep can speak their observations while still at the location — for example, 'Retailer mentioned a competitor is offering a 10% discount on their fungicide this month.'

Fully offline capable

The outcome log works with zero connectivity. Logged visits are stored locally and sync to the server automatically when the rep's phone reconnects.

Return to route

After submitting, the rep is taken back to the Route Planning screen with the completed stop marked and the next stop highlighted.

The Alerts Feed surfaces unusual patterns in the rep's territory that need immediate field action. It answers the question: what has changed since yesterday that I need to respond to today? Reps check this every morning alongside the dashboard.

Stockout cluster alert

Fires when three or more retailers in the same tehsil are out of stock on the same SKU. The alert lists all affected retailers, the product, and the number of farmers at risk. A direct action button lets the rep add all affected retailers to today's route immediately.

Demand spike alert

Fires when a SKU's weekly sales are more than double the prior week average. This often signals an emerging pest or disease problem — farmers are buying reactively. The rep needs to visit retailers in that tehsil before stock runs out and farmers turn to competitors.

Farmer product scan spike

Fires when multiple farmers in a tehsil scan the same Syngenta product within a short window. This is a demand intent signal — they are researching the product before buying. The rep should act before those farmers buy from a competitor.

Campaign heat alert

Fires when a digital campaign's impressions or clicks surge in a specific geography. The rep in that territory is prompted to reinforce the campaign in the field that week — visiting the relevant retailers while demand is warm.

Competitor activity pattern

Fires when reps across a territory have logged the same competitor product at multiple retailers in a short period. This surfaces as a territory-level competitive threat that may require a defensive response — pricing, promotions, or accelerated visits.

Weather-triggered alert

Fires when a weather event — unexpected rain, drought, or temperature spike — may shift the crop calendar. A planned fungicide campaign might become premature or, conversely, more urgent depending on the event.

Emergency mark

The rep can long-press any alert to escalate it to emergency status. This bumps the relevant visit to the top of the day's route and sends an automatic notification to the territory manager.

Data evidence on every alert

Every alert shows the specific data that triggered it — for example, '4 retailers with zero units of Actara 25 WG in Baramati tehsil as of this week.' Reps do not have to trust the alert blindly — they can see exactly why it fired.

Filter tabs

The feed can be filtered by category — All, Urgent, Opportunities, Campaigns, or Competitor. Alerts persist until the rep takes an action or logs a visit to the flagged retailer — they do not disappear just because the rep viewed them.

The Crop Scanner allows the rep to photograph a crop — a leaf, stem, or section of field — and receive an AI diagnosis of the disease, pest, or nutrient deficiency visible in the image. The diagnosis then connects directly to the AI chatbot, which recommends the best-fit Syngenta product. This is not a dead-end results screen — it starts a conversation.

Crop type selector

Before scanning, the rep selects which crop is being photographed — Wheat, Mustard, Chickpea, or Potato. This gives the AI critical context before it analyses the image, significantly improving diagnosis accuracy.

Camera capture

The rep photographs a leaf, stem, fruit, or affected area. The interface guides them to fill the frame and hold steady. Multiple photos can be taken for better accuracy.

AI disease diagnosis

The image is analysed and the diagnosis is returned in plain language — for example, 'Early-stage powdery mildew detected on wheat leaves. Severity: Moderate. This typically spreads rapidly during heading stage if untreated.' No technical jargon. The rep can read this out directly to the farmer.

Severity rating

Every diagnosis includes a severity level — Mild, Moderate, or Severe — and a confidence score so the rep knows how certain the AI is.

Chatbot continuation

After the diagnosis, the AI chatbot opens automatically with full context loaded. It continues: 'Based on this moderate powdery mildew, here is what I recommend and why.' The rep can ask follow-up questions, get dosage guidance, and discuss application timing.

Syngenta product recommendation

The chatbot recommends the most appropriate Syngenta product with dosage guidance and application timing based on the current crop growth stage and severity of the problem.

Yield loss comparison

Side-by-side projection of expected yield loss without treatment versus with the recommended Syngenta product — shown in kilograms per acre and rupee value. This is the rep's economic argument to the farmer.

Shareable advisory

One tap creates a WhatsApp-ready message containing the diagnosis, product recommendation, dosage, and yield comparison. The rep sends this to the farmer's phone directly.

Saves to farmer profile

If the rep arrived at this screen through a farmer's profile, the scan is automatically saved to that farmer's record with the date, photo, diagnosis, and recommendation.

Offline capability

A lightweight on-device AI model handles basic diagnosis for the four main Rabi crops without connectivity. Full analysis — including the chatbot conversation — requires a connection and runs when the rep reconnects.

The AI Chatbot is the rep's always-available field co-pilot. It is accessible as a floating button from every screen in the app. It is powered by GraphRAG — meaning it reasons across the full knowledge graph of the rep's territory, not just a static FAQ.

Natural language questions

The rep types or speaks a question in plain English, Hindi, or Marathi and gets a specific, grounded answer. Questions like 'What should I pitch at my next stop?', 'Which tehsil has the most stockout risk this week?', or 'What is the right dose of Topik for wheat at heading stage?' all return precise, data-backed answers.

Voice-first mode

The rep can activate the chatbot by voice while driving between visits. They ask their question hands-free and the response is read aloud. This is the most practical feature for a rep who spends hours driving through rural territory every day.

Product recommender

Given a crop type, growth stage, and observed problem, the chatbot recommends the most appropriate Syngenta product with the reasoning behind the choice — not just a product name.

Agronomic advisor

The chatbot answers crop-specific agronomic questions grounded in current growth stage, local weather, and pest risk for the rep's territory. It can advise on application timing, mixing instructions, and what to watch for in the coming week.

Territory intelligence queries

Because the chatbot uses GraphRAG to traverse the full territory knowledge graph, it can answer complex multi-hop questions like 'Which retailers in Shirur are understocked on the product I should be promoting for the wheat campaign this week?' — a question that requires joining retailer inventory, campaign data, territory data, and crop calendar data simultaneously.

Post-scan integration

After a Crop Scanner session, the chatbot continues the conversation with full diagnosis context loaded. The rep can relay questions from the farmer and get immediate answers.

Conversation history

The full day's conversation is saved and visible to the rep. They can scroll back to recall a recommendation from earlier in the day.

Multilingual responses

The chatbot responds in the rep's preferred language — Hindi, Marathi, or English — set once in the profile settings.

Partial offline capability

Without connectivity, the chatbot can answer from a cache of agronomic advice and Syngenta product information. Live territory queries — which require the GraphRAG database — are queued and answered when the rep reconnects.

The Reasoning Graph is the explainability screen behind every AI recommendation. When the rep (or the chatbot) suggests a product, the rep can tap 'Why this recommendation?' and see the exact evidence graph the AI used — which signals were considered, how they connect, and what logical steps produced the output. It is the most transparent view of AI reasoning available in any comparable agri-sales tool.

Example shown in the prototype: Farmer Ramesh, wheat variety HD-2967 at flowering stage. The AI recommends Tilt 25EC fungicide. Six signals feed the recommendation: the farmer's crop variety and stage, a 48mm rainfall event, 89% humidity sustained for 18 hours, a matching 2023 outbreak pattern in the same panchayat, and confirmed stock 5km away.

The Graph Visualization

Center recommendation node

A dark green node at the center of the canvas shows the recommended product — 'Tilt 25EC — Recommendation'. This is the output. Every other node on the canvas feeds into it via a directed edge. The center node pulses with a breathing glow animation to draw the eye and signal that it is the conclusion node.

Six signal nodes around the perimeter

Six nodes orbit the center, each representing a distinct evidence type. Each edge carries a verb label describing the relationship: 'treats' (farmer to product), 'for' (crop variety to product), 'triggers' (rainfall to risk), 'amplifies' (humidity to risk), 'matches' (historical outbreak to pattern), 'available at' (nearby stock to logistics). The label makes the causal chain readable at a glance.

Animated traveling pulse

A small amber dot travels along each edge toward the center node in a looping animation, visualising the flow of evidence into the recommendation. Each edge has a staggered delay so pulses arrive at the center in sequence rather than simultaneously.

Dim-on-select

Tapping any outer node dims all other nodes and edges to near-invisible. The selected signal and its connection to the center are highlighted in full opacity. This lets the rep focus on one piece of evidence at a time when explaining to a farmer or retailer.

Legend badge

A pill badge in the top-right corner reads '6 signals · 1 outcome' so the rep immediately knows the scale of evidence behind the recommendation without reading every node label.

The Six Evidence Nodes — Detail

Node	Type	Key Evidence
Ramesh	Person	Farmer profile · Member since 2021 · Last visited 7 days ago
HD-2967	Crop Variety	Wheat · Moderate Septoria susceptibility · 3.2 acres · Flowering stage (BBCH 65)
48mm rain	Environmental Signal	Mon–Tue window · 31h saturation · Exceeds fungal risk threshold by 12mm
89% humidity	Environmental Signal	18 hours sustained · Canopy wetness at dawn · Elevated spore release
2023 outbreak	Historical Match	Same panchayat (Bhatpura + Mallawan) · 12–22% yield loss · 0.87 cosine similarity
Stock 5km	Logistics Signal	Kisan Store, Sandila Rd · Tilt 25EC 500ml × 14 units · 11 min drive

Node Detail Flyout

Bottom sheet on tap

Tapping any outer node slides up a bottom sheet showing the node's type label (e.g. 'ENVIRONMENTAL SIGNAL'), its name, and a structured fact table of the specific values that triggered its inclusion in the graph. For example, tapping '48mm rain' reveals the exact monitoring window, the source weather stations (Hardoi block), and the precise threshold exceedance value (12mm above fungal risk threshold).

Data provenance link

Every flyout shows a 'Source' attribution line with an external link icon — for example 'Source · IMD station feed' or 'Source · IARI cultivar database'. The rep can verify the raw data behind any signal. This is critical for credibility when the rep is explaining the recommendation to a sceptical farmer.

Scrim and dismiss

Tapping outside the flyout or the close button dismisses it and returns the graph to its neutral full-opacity state. The flyout uses a blurred scrim behind it so the graph remains visible in the background.

Reasoning Trail

Below the graph canvas, a numbered reasoning trail shows the five logical steps the AI took in sequence — not just what signals were used, but in what order the inference was constructed:

Step	Inference	Source
1	Detected wheat at flowering stage from farmer profile	Farmer profile
2	Connected rainfall and humidity to fungal risk model	IMD + risk engine
3	Matched current pattern against 2023 outbreak in same panchayat	Outbreak ledger
4	Filtered Syngenta products for early-stage blight efficacy	Product catalogue
5	Identified in-stock location within 5km radius	Retailer inventory

Action Buttons

Use this answer

A primary CTA button at the bottom of the screen lets the rep accept the recommendation and return to the AI Chatbot or Visit Copilot with the product pre-selected. The reasoning graph is a drill-down, not a dead end — it feeds back into the action flow.

Ask follow-up

A secondary link opens the chatbot with the current recommendation loaded as context. The rep can ask 'Is there a cheaper alternative?' or 'What is the exact dose per acre?' and the chatbot responds with the full graph context already in its working memory.

Share

A share button in the top-right header lets the rep export the reasoning graph as a screenshot or structured summary — useful when reporting back to a manager or documenting why a specific recommendation was made on a specific farm visit.

The Yield Loss Calculator gives the rep a data-backed economic argument for the farmer. Instead of asking the farmer to trust a product, the rep shows them in rupees exactly how much they stand to lose by not treating, versus the cost of the Syngenta product.

Crop and farm size input

The rep enters the crop type and the farmer's farm size in acres. Farm size data pre-fills automatically if the rep is accessing this calculator from a farmer's profile.

Problem type and severity

The rep selects the type of infestation or disease — weed, fungal disease, or insect pest — and the severity level based on what they have observed or diagnosed via the Crop Scanner.

Without-product projection

Shows the expected yield loss in kilograms per acre and total kilograms for this farm, and converts that to a rupee loss using current market crop prices.

With-Syngenta-product projection

Shows the expected yield protection percentage with the recommended product, the net yield saved, the cost of the product, and the net return on investment.

Break-even calculation

Shows how many kilograms of yield the farmer needs to save to recover the cost of the product. This is often the most persuasive number — for most farmers it is very small relative to the total expected loss.

Live market price input

The rep can update the crop price if they know the local mandi rate. This makes the calculation immediately relevant to that farmer's specific situation.

Pre-loaded product costs

Prices for Topik 15 WP, Score 250 EC, Actara 25 WG, and Kavach 75 WP are pre-loaded so the rep does not need to remember or look them up.

Shareable output

The calculator generates a simple, clean summary — numbers and a brief explanation — that the rep can show on screen or send to the farmer via WhatsApp.



Farmer profiles are created and managed entirely by the field rep. There is no separate login or access for farmers. The rep is the relationship owner, and the profile reflects everything the rep knows about each farmer in their territory.

Basic demographics

Farmer name (rep-entered), tehsil, district, state, age, gender, and primary language. Language is important for the rep to know — it determines which language to use in conversation and which advisory materials to share.

Device type

Whether the farmer has a smartphone, a keypad phone, or unknown. This determines whether the farmer can receive WhatsApp messages or only voice calls.

Farm size

Farm size in acres. Used by the Yield Loss Calculator to compute total rupee impact.

Crop calendar

A visual timeline of the farmer's Rabi crops — Wheat, Mustard, Chickpea, or Potato — with key biological stages marked and the current stage highlighted. This is the primary driver of visit timing recommendations for this farmer.

Purchase history

Every Syngenta product this farmer has bought, when they bought it, and from which retailer. Helps the rep understand the farmer's buying patterns and loyalty.

Offline campaign attendance

Which grower meetings or field events this farmer has attended and when. Farmers who attended an event recently are warmer prospects and should be followed up.

Crop scan history

Every diagnosis done for this farmer via the Crop Scanner — saved with the date, photograph, diagnosis result, and product recommendation. Gives the rep a history of problems this farmer has faced.

Visit log

All rep visits where this farmer was discussed or visited directly, with outcomes and notes. The rep can see the full relationship history in one scrollable view.

Rep notes

A free-text field updated after each interaction. Voice-to-text input available. Notes are timestamped so the rep can see what changed over time.

Quick actions

One tap to navigate to this farmer's nearest retailer, open the chatbot with this farmer's profile pre-loaded as context, or open the Yield Loss Calculator pre-filled with this farmer's crop and farm size.

Retailer profiles are created and managed by the field rep. No retailer login exists. The profile aggregates everything relevant about a retail outlet — stock history, visit log, nearby farmers, and competitor observations — in one place.

Basic information

Retailer name, owner name, tehsil, district, state, and the best time to visit based on the rep's historical notes — for example, 'Avoid Mondays, owner at market.'

Current stock status

Live inventory for all key SKUs — Out of Stock shown in red, low stock in amber, healthy in green — with exact unit counts from the most recent weekly snapshot.

Stock history as sparklines

A small chart showing weekly inventory levels per SKU over the past several weeks. The rep can see the drawdown rate and predict when the retailer will next run out.

Sales history

What has sold from this retailer, in what quantities, and over what time period. The rep can identify which products move fast here and which do not.

Visit log

Full history of rep visits — date, what was pitched, outcome logged, and notes. The rep can see at a glance that the last three visits all resulted in no purchase and adjust their approach accordingly.

Competitor products

A log of competitor SKUs the rep has observed stocked at this retailer, with the date each was observed. This builds the territory-level competitor intelligence picture that the manager sees in their console.

Nearby farmers

List of farmers in this tehsil who are likely customers of this retailer, shown with their crop type and current growth stage. The rep uses this to understand what products the retailer's customers are likely to need right now.

Rep notes

Free-text field with timestamps. Voice-to-text available. The rep can log practical observations — 'Owner is interested in Actara bulk deal for next season' — and retrieve them at the next visit.

Quick actions

One tap to start navigation to this retailer, open the Visit Copilot for this outlet, or open the chatbot with this retailer's context pre-loaded.

Offline Mode is not a separate screen — it is a system-wide layer that runs underneath the entire app. Rural India has unreliable connectivity and any feature that breaks without signal is a feature that fails on the job. Every critical action in AgroPilot works offline.

The offline architecture is explicitly required by the problem statement. It is also a practical differentiator — most competing solutions will be purely cloud-dependent and will fail in the field.

Morning pre-cache

When the rep opens the app each morning with connectivity, it automatically downloads everything needed for the day — all visit cards, farmer profiles, retailer profiles, territory map tiles, AI recommendations, and alert data. By the time the rep is in their vehicle, the app is ready to work offline.

Offline visit logging

Logging a visit outcome, adding rep notes, scanning a crop, or editing a profile all work without connectivity. Every action is stored locally in a sync queue.

Offline crop scanning

A lightweight on-device AI model handles basic disease diagnosis for Wheat, Mustard, Chickpea, and Potato without an internet connection. Full analysis and chatbot continuation run when connectivity returns.

Offline map navigation

Territory map tiles are cached each morning. The rep can view the map, see all visit pins, and follow the route without connectivity.

Offline chatbot

A cache of agronomic advice and Syngenta product information allows the chatbot to answer common questions without connectivity. Live territory queries are queued and answered automatically when the phone reconnects.

Sync queue indicator

The offline banner in the top bar shows the number of items waiting to be synced — for example, '3 visit logs pending upload.' The rep always knows what has not yet reached the server.

Auto-sync on reconnect

When the phone reconnects — even briefly — the sync queue uploads automatically in the background without interrupting the rep.

Conflict resolution

If the same profile was edited both offline and online simultaneously — for example, a manager updated a retailer record while the rep was offline — the latest timestamp wins and the rep is shown a brief notification about what changed.

Data saver mode

In very low bandwidth situations, the rep can enable Data Saver mode — which disables map tile loading and reduces image quality in the Crop Scanner to minimise data usage.

Offline status banner

A persistent amber banner at the top of every screen indicates when the app is running offline. It disappears automatically when connectivity returns and sync completes.

Manager Console

The Manager Console is a separate web application accessed via desktop or laptop browser. Managers log in with a different role from reps. The console shows aggregated data across all reps and territories — it is a command centre for regional oversight, not a field tool.



The Territory Heatmap gives the manager a visual, geographic view of everything happening across all territories. Gaps, hotspots, and anomalies are immediately visible without reading through lists of numbers.

Visit coverage heatmap

Each tehsil is colour-coded by visit coverage percentage — how many retailers and farmers in that tehsil were visited in the current week. Dark green means well covered. Red means the tehsil has had no visits recently and may be at risk of stockouts or competitor capture going undetected.

Demand signal overlay

A toggleable layer shows demand spikes and stockout clusters as map indicators. The manager can immediately see where the urgent inventory problems are geographically.

Competitor activity layer

A toggleable layer shows where reps have logged competitor product sightings. Clusters of competitor activity in one tehsil are visible as a map hotspot.

Rep activity trails

The manager can select any rep and see their visit path for the current or previous week — which retailers were visited, in what order, and what outcomes were logged.

Tehsil drill-down

Clicking any tehsil on the map opens a panel showing all retailers and farmers in that tehsil, the rep who covers them, the last visit date, and current stock status for key SKUs.

Filters

The map can be filtered by date range, individual rep, product, crop, or alert type. The manager can isolate, for example, all tehsils where Actara 25 WG is out of stock to see the geographic scope of the problem.

Cold zone automatic highlighting

Tehsils with zero visits in the past two weeks are automatically highlighted with a warning indicator. The manager does not need to manually find neglected areas — they surface automatically.

The KPI Dashboard tracks the three success metrics named explicitly in the problem statement: revenue per field day, coverage efficiency, and recommendation acceptance rate. All metrics are shown with week-on-week trends, not just point-in-time numbers.

These are the exact metrics Syngenta defined as 'what successful outcomes look like' in the problem statement. The dashboard is designed to make these measurable from day one of the prototype.

Revenue per field day

Total sales value generated divided by the number of days the rep was active in the field — shown per rep, per territory, and as a regional aggregate, all with week-on-week delta and target comparison.

Coverage efficiency

The percentage of retailers and farmers in each territory that were visited in the current week. Broken down by rep and by tehsil so the manager can see not just whether coverage targets were met but where the gaps are.

Recommendation acceptance rate

How often the rep's AI-suggested pitch — the product the AI recommended leading with on the Visit Copilot screen — resulted in a sale or order. Tracked per rep, per product, and over time. A declining acceptance rate may signal that AI recommendations are not well-calibrated for a particular territory.

Visit-to-sale conversion rate

Across all logged visits, what percentage resulted in a sale or order. Broken down by product, tehsil, rep, and week. Identifies which product pitches are converting and which are not.

Digital-to-field funnel linkage

Tracks whether reps visited retailers in tehsils where the digital campaign showed high engagement that week. This measures whether digital and field activities are coordinated or operating in isolation.

Week-on-week trend lines

All KPIs are shown with trend lines over the past eight weeks, not just the current week's numbers. The manager can see whether performance is improving, declining, or plateauing.

Export to CSV

All dashboard data can be exported for use in external reporting or presentations.

The Rep Performance Tracker gives the manager individual scorecards for every field rep. It goes beyond simple revenue numbers — it tracks the quality of field activity, including whether reps are using the AI system effectively.

Individual rep scorecards

Each rep has a scorecard showing their visits completed, outcomes logged, coverage achieved, revenue generated, and recommendation acceptance rate for the current week and the trailing four weeks.

Leaderboard view

The manager can rank all reps by any single metric — revenue per field day, coverage percentage, or recommendation acceptance rate. This surfaces both top performers and reps who may need coaching.

AI recommendation usage rate

Tracks whether each rep is following the AI's visit priority list and product recommendations, or ignoring them. A rep with a high acceptance rate but low AI usage rate may be performing well despite the system — or may be a data quality problem if they are not logging outcomes properly.

Visit quality score

A composite score based on outcome logging completeness, follow-up rate, and whether visits align with the AI priority list. A rep who visits many retailers but logs no outcomes has a low quality score even if their visit count is high.

AI-flagged underperformance

The system automatically flags reps whose territory shows patterns of concern — ignoring urgent alerts, leaving high-priority retailers unvisited for extended periods, or consistently low recommendation acceptance rates — so the manager can intervene before the territory is significantly impacted.

Emergency response time

Tracks how fast each rep acts after a high-severity alert is raised. A rep who consistently responds to emergencies within the same day is more field-effective than one who lets alerts sit for a week.

The manager's Alert Management screen aggregates every anomaly and opportunity detected across all territories into a single feed. The manager sees the same alerts as reps — and can add their own context, assign actions, and track resolution.

Cross-territory alert feed

All anomalies — stockout clusters, demand spikes, competitor activity patterns, weather triggers — from every rep's territory in one consolidated view. The manager can see regional patterns that no individual rep can see from their own territory alone.

Emergency escalation inbox

Alerts that reps have long-pressed and escalated to emergency status appear here immediately, with the rep's name, location, and what they flagged. The manager can respond with a note or phone call within seconds.

Assign to rep

The manager can push any alert directly to a specific rep's queue with a note attached. The rep sees the alert and the note in their Alerts Feed.

Resolution tracking

Every alert is tracked through to resolution — was the relevant retailer visited after the alert fired? Was an outcome logged? Did stock levels recover? The manager can see the full lifecycle of each alert.

Competitor intelligence summary

A summary view of all competitor product sightings across the region — which brands are appearing in which tehsils, at what frequency, and whether the pattern is intensifying or stable over time.

Resolution rate metric

What percentage of alerts resulted in a confirmed field action within 48 hours. This measures how responsive the field force is to the AI's anomaly detection — a key indicator of whether the system is actually changing behaviour.

The Campaign Performance screen links digital marketing activity to field sales outcomes. It answers the question that neither the digital team nor the field team can answer alone: did the digital campaign actually drive sales in the field?

The four active Rabi campaigns:

Campaign ID	Crop	Product
CMP_RABI25_001	Wheat	Topik 15 WP
CMP_RABI25_002	Mustard	Score 250 EC
CMP_RABI25_003	Chickpea	Actara 25 WG
CMP_RABI25_004	Potato	Kavach 75 WP

Digital funnel per campaign

For each of the four campaigns — Wheat/Topik, Mustard/Score, Chickpea/Actara, Potato/Kavach — the manager sees week-by-week impressions, landing page visits, and lead form submissions. The full funnel from awareness to expressed interest.

Field reinforcement rate

In tehsils where a campaign's digital engagement surged in a given week, did the relevant rep visit retailers selling that product in the same week? This measures whether field and digital activities are actually coordinated.

Campaign-to-sale attribution

Did product sales in a tehsil increase the week after a campaign showed high engagement there? This is the critical attribution link — connecting a digital impression to a physical sale through the rep's field visit.

Week-on-week campaign trends

All four campaign funnels shown as weekly trend lines so the manager can see which campaigns are gaining momentum and which are plateauing.

Underperforming campaign flags

If a campaign is generating high impressions but low visits and near-zero leads, the system flags it as a landing page or targeting problem — not a field problem. If impressions are low but field sales are high in that tehsil, the campaign budget may need to be reallocated there.

Technology Stack

The technology choices are deliberate and each serves a specific purpose. The core differentiator is GraphRAG — the ability to answer multi-hop questions across interconnected datasets that no simple database query or traditional RAG system can handle.

Layer	Technology	Purpose
Knowledge Graph	Neo4j	Stores all relationships between reps, territories, retailers, farmers, SKUs, campaigns
Graph RAG	Microsoft GraphRAG + LlamaIndex	Retrieves context by traversing the knowledge graph rather than simple vector search
AI Reasoning	Claude 3.5 Sonnet (Anthropic)	Generates explainable recommendations, agronomic advice, and chatbot responses
Agent Orchestration	LangGraph	Multi-step AI agent: plan → query graph → fetch signals → score → reason → respond
Embeddings	OpenAI text-embedding-3-small	Semantic search over visit notes, product descriptions, and agronomic content
Backend API	FastAPI (Python)	REST endpoints for the mobile app and manager console
Database	PostgreSQL + Neo4j	Transactional data in PostgreSQL, relationship graph in Neo4j
Mobile App	React (PWA)	Progressive Web App — works offline via Service Workers, installable on Android
Manager Console	Next.js	Web dashboard, desktop-optimised
Maps	Mapbox GL JS	Territory maps with custom agricultural styling
External Signals	OpenWeatherMap, NASA MODIS	Tehsil-level weather and satellite NDVI crop health index
Offline Storage	IndexedDB + Service Worker	Client-side storage for offline visit logs, cached profiles, and queued sync
Infrastructure	Railway (backend) + Vercel (frontend)	Fast deployment for prototype; production-ready paths available

The Knowledge Graph — Why It Matters

Most solutions will connect to the datasets using SQL queries or simple vector search. AgroPilot uses a knowledge graph because the most valuable questions require traversing relationships across multiple tables simultaneously:

Example query 1

Find all retailers in this rep's territory where Topik 15 WP is out of stock, AND where nearby farmers have wheat crops in heading stage this week. This is a 4-hop graph traversal. It is impossible to write as a readable SQL join and returns instantly as a Neo4j Cypher query.

Example query 2

Which farmers in Shirur tehsil attended the offline campaign AND have not yet bought Actara 25 WG AND whose chickpea crop enters the pod-fill stage this week? This is the exact kind of precision targeting that GraphRAG makes effortless.

Example query 3

Which retailers are serving farmers who clicked the WhatsApp campaign but have not yet been visited by the rep since the click? This identifies the highest-value follow-up actions from digital activity.

How Each Dataset Powers the Features

The application uses all eight datasets provided for the hackathon. Below is a plain-language explanation of what each dataset does in the product.

reps_territory.csv	500 rows	Defines each rep's territory and which tehsils they cover. This is the starting node in every GraphRAG query — all recommendations are filtered to this rep's jurisdiction first.
retailers.csv	4,000 rows	Geographic anchor for all retailers. The tehsil field is used to cluster retailers for route optimisation and to link retailers to nearby farmers.
retailer_visit_log.csv	30,000 rows	Two uses: visit_date tells the system how many days ago each retailer was last visited (the coverage gap score), and product_recommended prevents the AI from suggesting the same pitch that was made on the last visit.
retailer_inventory_weekly.csv	310,544 rows	The richest signal in the dataset. sku_qty = 0 triggers stockout alerts. Week-on-week comparison detects drawdown rate — if stock drops 40% in one week, a visit is recommended before the retailer goes out of stock.
retailer_pos.csv	235,042 rows	Demand signal and outcome verification. Week-on-week sales spikes trigger demand alerts. Also used after a rep visit to verify whether the pitch resulted in a sale — this closes the AI learning loop.
growers.csv	6,000 rows	The grower_crop_calendar JSON field is the crop stage timing engine — it drives the biological urgency component of the priority score. product_scan detects demand intent. offline_campaign_attended identifies warm follow-up opportunities.
digital_funnel_weekly.csv	104 rows	Campaign heat signal. Rising landing page visits on a product = digital demand warming up → rep should reinforce in the field this week, not next month.
whatsapp_message_log.csv	4,479 rows	clicked_status = True is the strongest individual farmer signal in the dataset. Clicks are aggregated by tehsil to create a demand warmth score — retailers serving tehsils with high click density are prioritised for visits.

SUMMARY

What Makes AgroPilot Different

What most solutions will do	What AgroPilot does
SQL queries + bar charts	Knowledge graph traversal with multi-hop reasoning
Static daily plan	LangGraph agent that re-plans dynamically as alerts fire
Generic product suggestions	Explainable AI pitch grounded in stock, crop stage, and weather
Online only	Full offline capability with morning pre-cache and auto-sync
Single LLM call	Multi-step agent: plan → query → signal → score → reason → output
Dashboard for managers only	Dual-interface: mobile co-pilot for reps + analytics console for managers
Guess-based prioritisation	Priority score from five weighted real data signals per retailer
Crop advice from FAQs	Live crop scanner + GraphRAG-powered agronomic chatbot

Page Count Summary

Part	Screens	Primary User	Platform
Part A — Field Rep App	12 screens	Field Sales Representative	Android Mobile (PWA)
Part B — Manager Console	5 screens	Territory Manager	Desktop Web Browser
Total	17 screens	—	—

This document was prepared for the Syngenta IITM Hackathon 2026 prototype submission. All dataset references are to the synthetic datasets provided on 13 May 2026. Contact: Ayush.Verma@syngenta.com for dataset queries.